

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO2_AT3 — CASE STUDY

Course Code:	DSA06	Course Title:	Data Handling and Visualization
CO Assessed:	CO2 – Evaluation (BL5)	Assessment:	Assessment Tool 3 – Case Study
Weightage:	20% of CIA	Total Marks:	20 (2 Questions × 10 Marks)
CO2 Statement:	Analyze real-world data scenarios and evaluate/justify the most appropriate chart types, coordinate systems, color scales, and dashboard designs to represent them.(BL5)		
Student Name:	Sania Herbert	Reg. No.:	192324062
Section / Batch:	2023/AI&DS	Date:	06.08.26

Code of Conduct

I Sania Herbert(192324062) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Sania

Instructions

- This test contains 2 case-study questions, each carrying 10 marks. Total: 20 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 90 minutes.
- Each case study has four sub-parts (a–d); answer all parts for full marks.
- Support your answers with brief calculations or justifications where relevant.
- Diagrams may be hand-drawn or clearly described in words if drawing tools are unavailable.

Annexure A – CASE STUDY

Rubrics for Calculation **ASSESSMENT RUBRIC**

AT3 — Case Study

CO2 · BT5: Evaluate ·

This rubric is used to assess student responses to the CO2-AT3 Case Study questions, in which students analyze a realistic multi-part scenario and select, justify, design, and critique appropriate visualizations across chart types, coordinate systems, color scales, and dashboard development.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough understanding of the full range of chart types, coordinate systems, and color scales relevant to the case, and correctly identifies which apply to each sub-part.	Good understanding of the relevant concepts, with only minor gaps in identifying appropriate chart types, scales, or color choices.	Basic understanding, but with some misconceptions about which visualization techniques suit the scenario's data.	Poor understanding of core concepts; proposes techniques fundamentally unsuited to the case study's data.
Explanation	25%	Provides detailed, well-reasoned justification for every design decision and critique, explicitly tied to specific details of the case.	Provides clear justification for most decisions and critiques, with adequate reasoning.	Justification is limited or generic; decisions are stated without sufficient reasoning tied to the case.	Little or no justification provided; answers are unsupported assertions.
Application	20%	Effectively applies visualization concepts to analyze and solve every sub-part of the case, including any calculations, with well-reasoned	Applies concepts correctly to most sub-parts, with only minor errors.	Limited application; some sub-parts are weakly addressed or only partially correct.	Fails to apply appropriate concepts; sub-parts are left unaddressed or answered incorrectly.

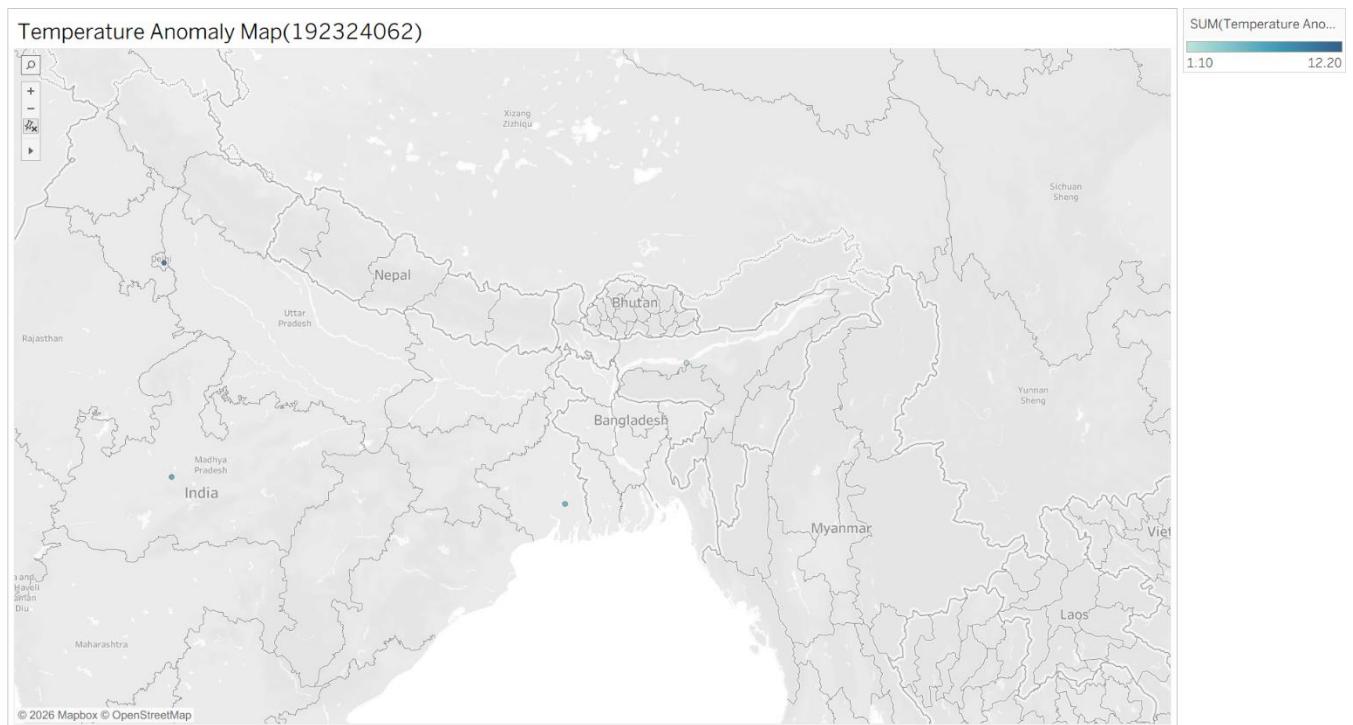
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
		conclusions.			
Organization	15%	The response is clearly structured, addressing every sub-part (a–d) in a logical sequence with coherent overall flow.	The response is organized and addresses each sub-part, with only minor issues in flow or clarity.	Basic structure; sub-parts are addressed but the response lacks clarity or a logical sequence.	Poorly organized; the response is incoherent, and sub-parts are left unaddressed or answered out of order.
Accuracy	10%	All parts of the answer — chart choices, calculations, critiques, and dashboard design — are completely accurate and well-suited to the case.	Mostly accurate, with only minor omissions or a small error in one sub-part.	Partially correct; some elements are missing, mismatched to the case, or incorrect.	Largely incorrect or inappropriate, with major gaps across multiple sub-parts.

Scoring Guide

For each criterion, award a performance level (1–4) based on the descriptors above. Multiply the level achieved (as a percentage of 4, i.e., Level 4 = 100%, Level 3 = 75%, Level 2 = 50%, Level 1 = 25%) by the criterion's weightage, then sum across all five criteria to obtain the total score out of 100%.

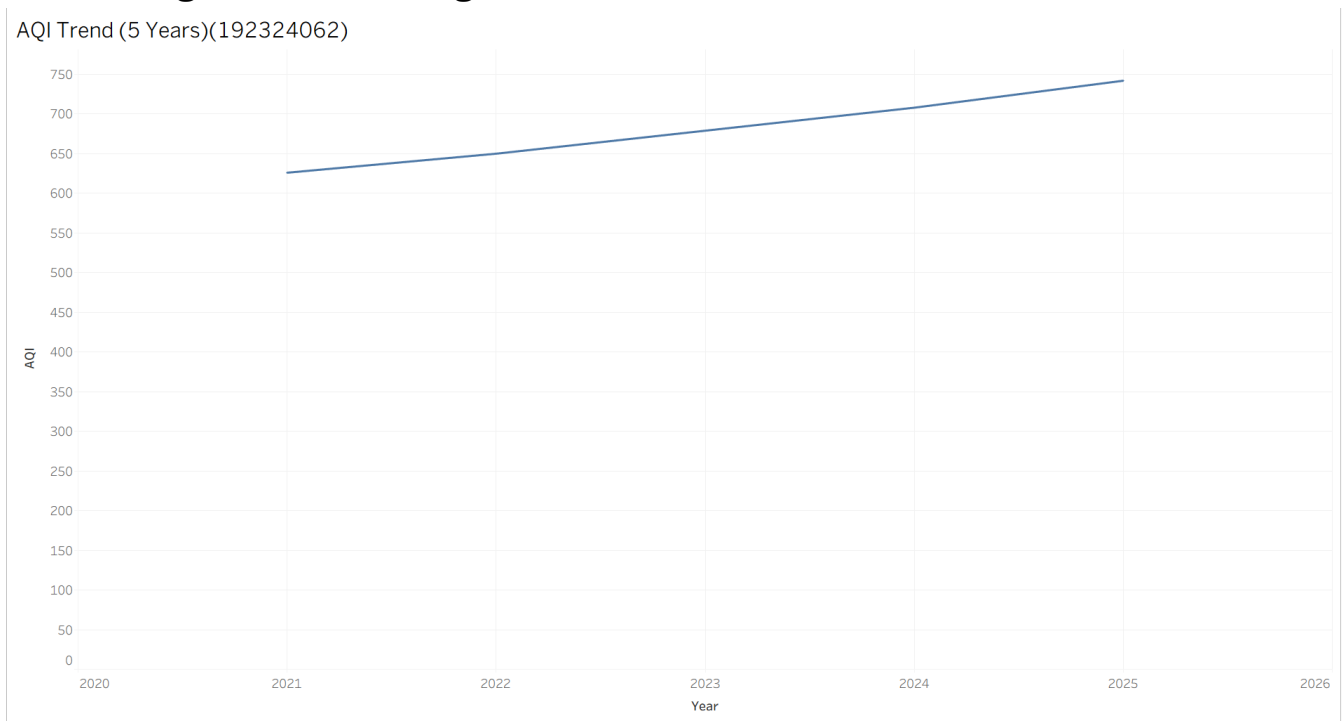
Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

QUE:5 a) What type of color scale is required for the temperature anomaly map, and why?



A diverging color scale highlights values above and below a meaningful midpoint (0°C anomaly), making warming and cooling regions easy to compare.

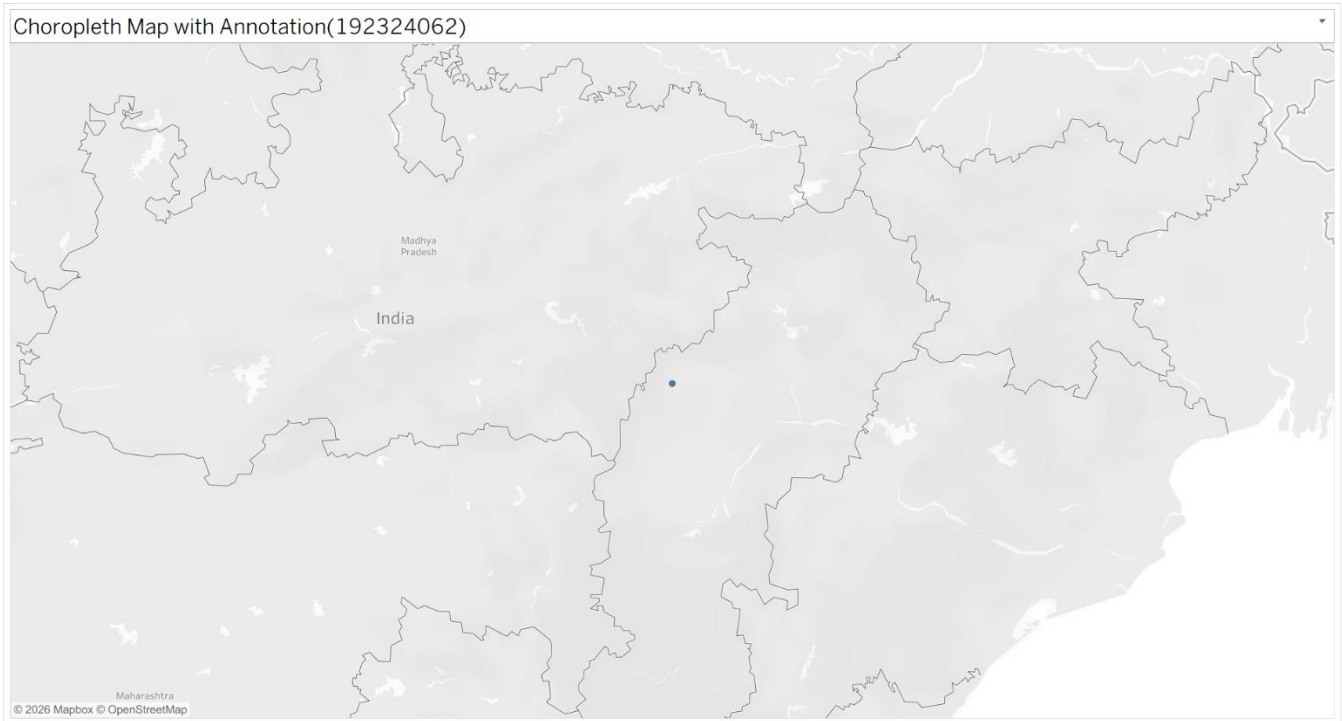
b) Propose a chart for the 5-year AQI trend, and explain how you would handle large seasonal swings in the data.



A line chart is ideal for showing trends over time, while a moving average reduces

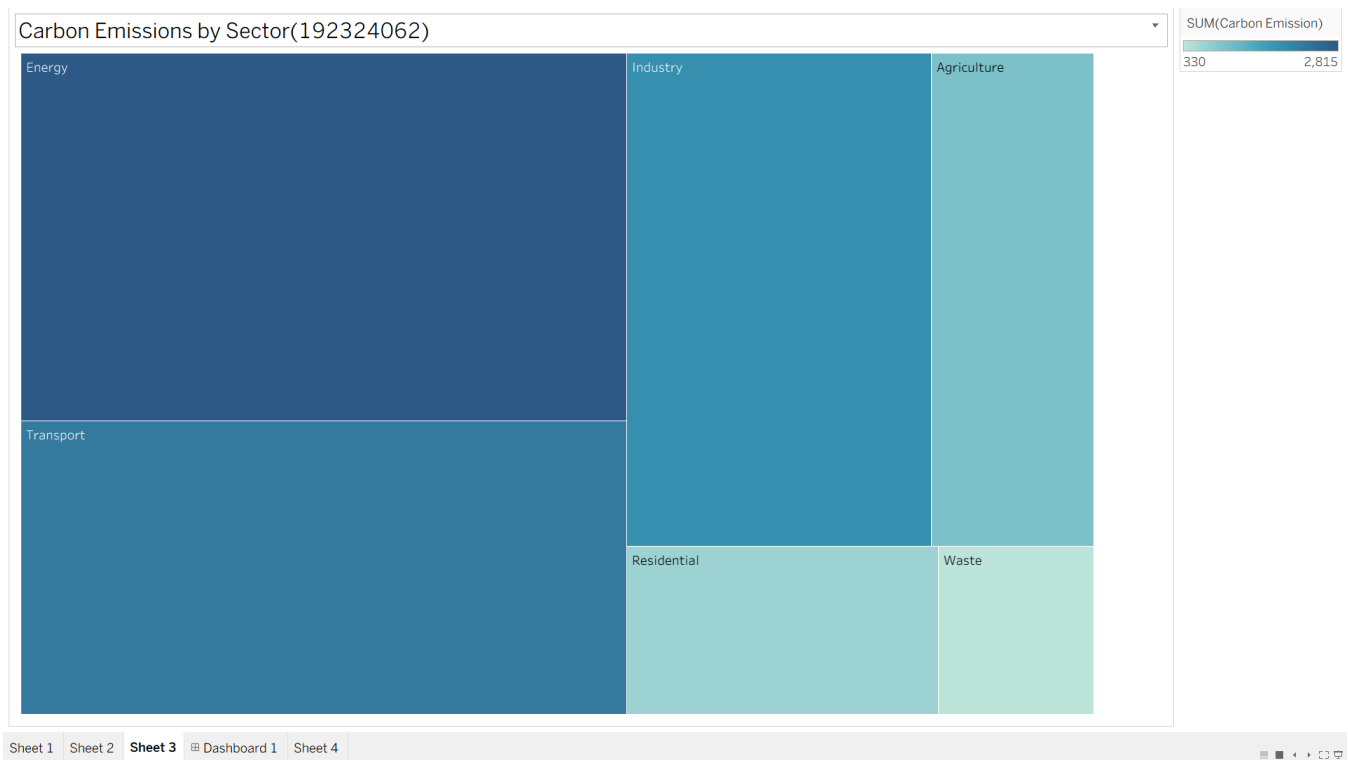
seasonal fluctuations and reveals the long-term pattern.

c) Design a choropleth-style map or grid for the temperature anomaly data, combined with an annotation on the most extreme region.



A choropleth map effectively displays spatial patterns in temperature anomalies, while an annotation immediately draws attention to the most extreme region.

d) Justify your choice of chart for carbon emissions by sector, and explain why a treemap might be preferred over a pie chart for this particular breakdown

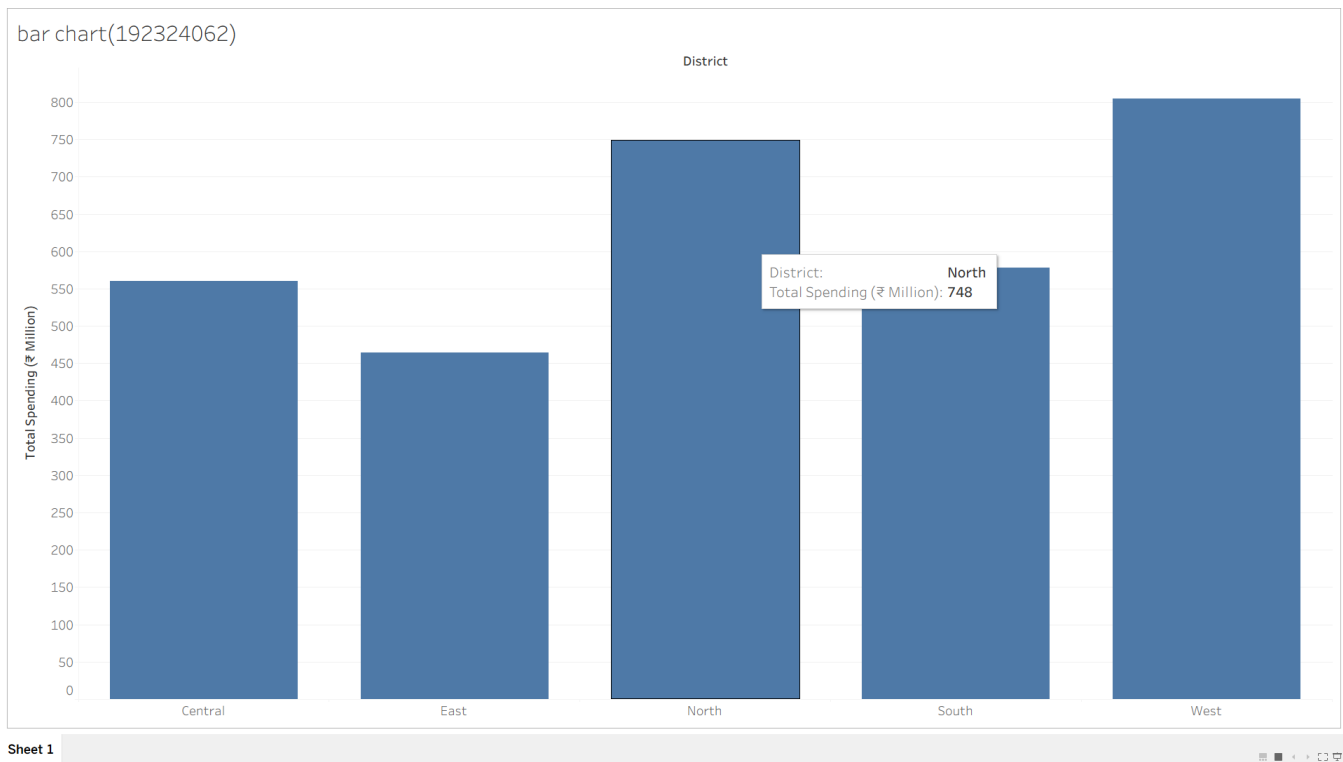


Why Treemap?

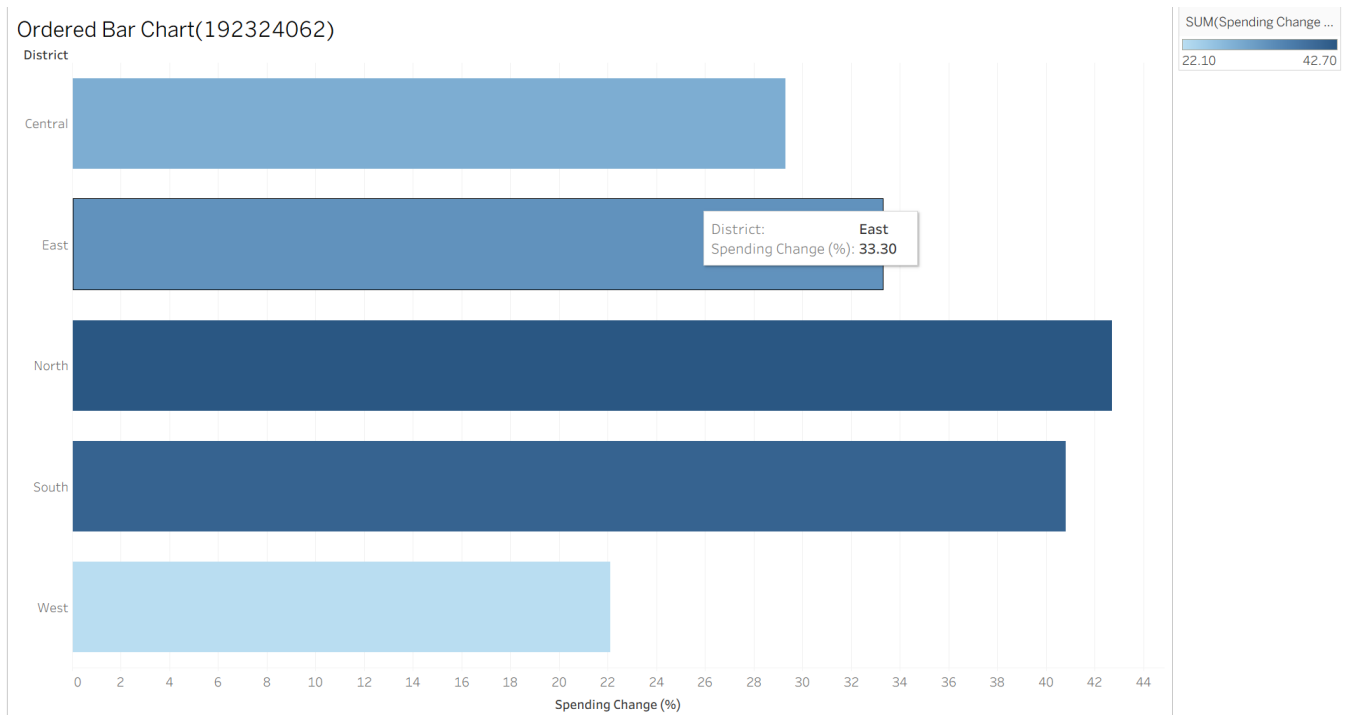
A treemap is better because:

- It displays many sectors clearly.
- Rectangle areas are easier to compare than pie slices.
- It efficiently uses space.
- Labels fit inside rectangles.
- Small categories remain visible.
- It scales well when there are many sectors.

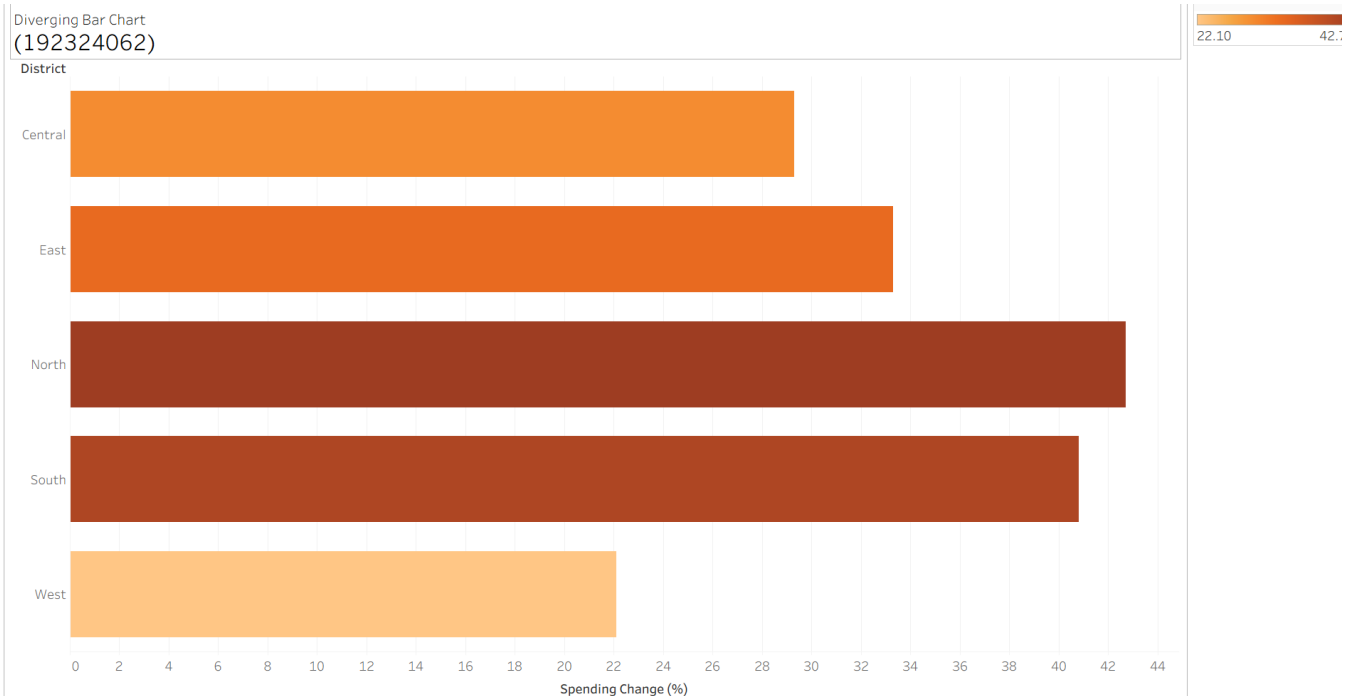
QUE:10 a) What axis scale is appropriate for the district spending data, and why?



b) Which chart correctly represents the ordinal service-satisfaction categories while preserving their order?

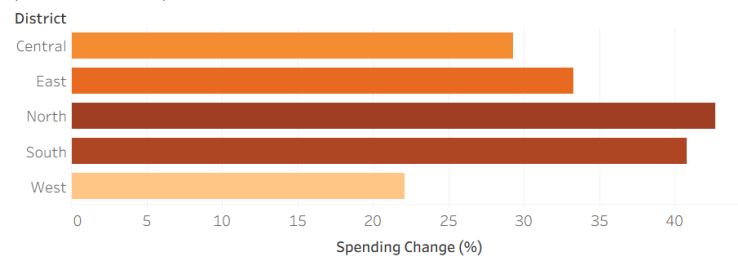


c) Propose a chart and color scale for the year-over-year spending change data (increases vs. decreases).

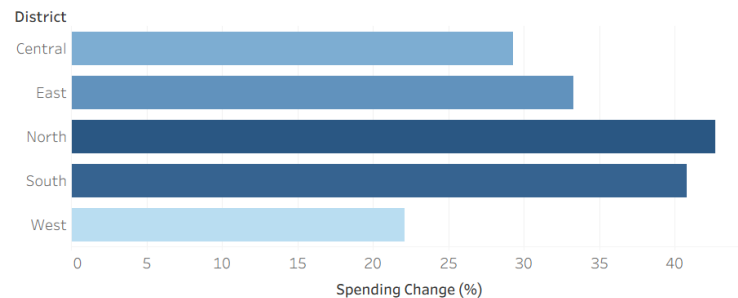


d) Design a combined dashboard using two of the above views, with a narrative takeaway a city council member could act on.

Diverging Bar Chart
(192324062)



Ordered Bar Chart(192324062)



bar chart(192324062)

